

Structural Timber Framing Construction Plan

12' x 18' Rectangular Cedar Pergola Package

Project Location: Saanich, BC, Canada (BC Coastal Jurisdiction)

Design Standard: BC Building Code 2024 / Part 9 Timber Framing

Ground Snow Load: 35.0 PSF | **Wind Exposure:** Category B

Structure Type: 6-Post Rectangular Post-and-Beam Pergola with Decorative Scallop Overhangs

Material Species: Heavy Timber Yellow Cedar / Western Red Cedar

1. Executive Summary & Design Rationale

This structural package provides complete fabrication specifications, load-path engineering, foundation schedules, and orthographic shop blueprints for a **12' x 18' rectangular heavy-timber pergola**. The layout features a symmetric 6-post grid (9' span along the 18' length and 12' clear transverse span) designed to accommodate zoned outdoor dining and lounge relaxation.

Core Dimensions & Specifications

- Footprint:** 18'-0" (Length) x 12'-0" (Width)
- Clear Height:** 8'-8" (9'-6" Top of Beams, 10'-0" Top of Rafters)
- Posts (P1-P6):** 6x6 Nominal (5.5" x 5.5" actual) Yellow Cedar, 9'-6" cut length
- Perimeter Beams (B1-B6):** 4x8 Nominal (3.5" x 7.25" actual) Yellow Cedar
- Primary Cross-Rafters (R1-R6):** 4x6 Nominal (3.5" x 5.5" actual) with classic decorative scallop profiles
- Lateral Knee Braces (K1A-K6B):** 4x4 Nominal (3.5" x 3.5" actual) paired 45° knee braces at all post-to-beam junctions
- Footings (FT1-FT6):** 10" diameter Sonotube concrete piers (minimum 36" depth below frost line) with Simpson Strong-Tie ABA66Z standoff bases
- Sun Sail Reinforcement:** Engineered tension tie points integrated into beam-post junctions

2. Photorealistic Architectural Visualization

Below is the approved architectural rendering illustrating the completed 12' x 18' structure in a landscaped West Coast garden setting:

 Photorealistic Pergola Render

3. Architectural Drawings (Presentation Layer)

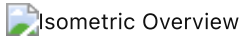
Sheet A01 — Architectural Plan View

 Plan View

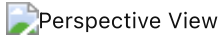
Sheet A02 — Architectural Elevation View

 Elevation View

Sheet A03 — Isometric Overview



Sheet A04 — Perspective Framing View



4. Orthographic Shop Blueprints (Fabrication Layer)

Sheet S01 — Engineering Foundation & Framing Plan



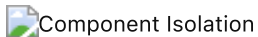
Sheet S02 — Structural Framing Elevation & Joint Miters



Sheet S03 — Structural Isometric Assembly



Sheet S04 — Component Details & Member Isolation



5. Bill of Materials & Cut Schedules

Refer to the accompanying standalone builder documentation for complete breakdown:

- **Cut List:** `outputs/fabrication/cut-list.json` & `outputs/shop-blueprint/SB01-cut-list.json`
- **Lumber Purchase Order:** `outputs/lumber-purchase-list.md`
- **Budget Estimate:** `outputs/budget-estimate.md`
- **Assembly Guide:** `outputs/assembly-guide.md`